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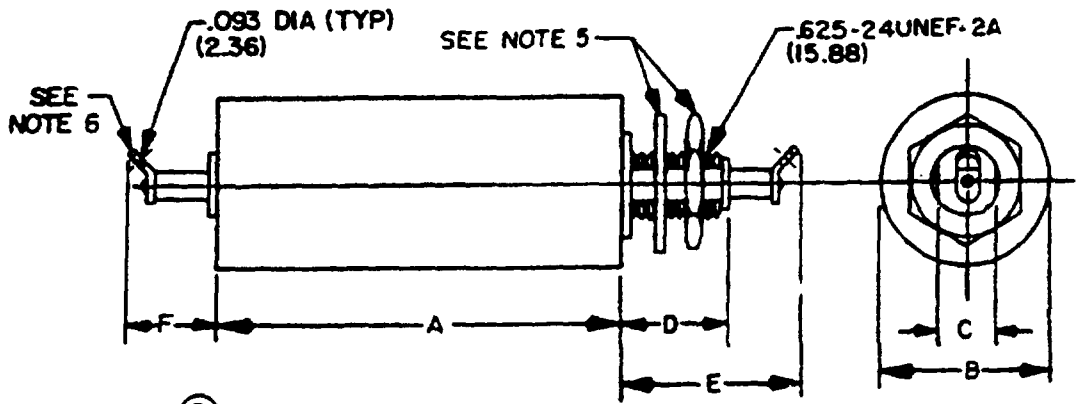
MIL-PRF-15733/55D
6 October 1983
SUPERSEDING
MIL-F-15733/55C
8 August 1979

PERFORMANCE SPECIFICATION SHEET

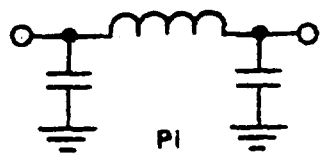
FILTERS, RADIO INTERFERENCE, HERMETICALLY SEALED
STYLE FL85

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.



Ⓓ DASH NO 0001, 0002 AND 0003



CIRCUIT CONFIGURATION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Circuit diagram is for information only.
5. Mounting hardware (internal-tooth lockwasher and hex nut) shall be supplied with filter.
6. Terminal shape and angular orientation optional.

FIGURE 1. Case dimensions and circuit configurations

Ⓓ denotes changes

FSC 5915

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TABLE I. Filter dimensions.

Type designation 1/ 2/ 3/	Dash number	A		B		C	
		Min	Max	Min	Max	Min	Max
FL85FW1EE3	0001	2.907 (73.84)	3.093 (78.56)	1.063 (27.00)	1.187 (30.15)	.526 (13.36)	.536 (13.61)
FL85FW1EE3	0002	2.907 (73.84)	3.093 (78.56)	1.063 (27.00)	1.187 (30.15)	---	.531 (13.49)
(D) FL85YW1EE3	0003	2.31 (58.7)	2.43 (61.7)	1.235 (31.37)	1.265 (32.13)	.365 (9.27)	.385 (9.78)

Dash number	D		E		F	
	Min	Max	Min	Max	Min	Max
0001	.359 (9.12)	.391 (9.93)	.625 (15.88)	.781 (19.84)	.282 (7.16)	.406 (10.31)
0002	.359 (9.12)	.391 (9.93)	.625 (15.88)	.781 (19.84)	.282 (7.16)	.406 (10.31)
(D) 0003	.47 (11.9)	.53 (13.5)	---	1.43 (36.32)	---	.50 (12.70)

1/ The letter "W" indicates insertion-loss versus frequency characteristics specified in tables I, II and III.

2/ The letter "E" indicates the operating temperature range -55°C to +85°C.

3/ The type designation does not describe a discrete item. For complete identification, the part number and type designation should be referenced.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Weight: 5.0 ounces, (0001, 0002).

Case: Metallic, hermetically sealed.

Terminals: Solderable (see figure 1).

Operating temperature range:

-55°C to +85°C; temperature characteristic E.

Rated voltage: 400 V dc or 125 V rms at 400 Hz over the rated temperature range of -55°C to +85°C (0001, 0002). 125 V ac at 400 Hz, 400 V dc (0003).

Rated current:

Amperes dc or ac (rms)	Dash nos.
5	0001
10	0002
2.0	0003

DC resistance: 0.072 ohm, maximum.

Insertion loss characteristic: Symbol W.

Seal: Filters shall be placed with the terminals facing sideways (not upward), and brought to a case temperature of 90°C ±5°C. After 10 minutes at this case temperature, the filters shall be turned through 180° onto another surface with the terminals still sideways and kept at this position for 10 minutes with the case temperature maintained within the same limits. Filters which show liquid seepage shall be cleaned and cooled to room temperature. While still at room ambient temperature, the filters shall be immersed for 1 minute in oil or water maintained at a temperature of 90°C ±5°C. The filters shall be immersed horizontally so that the terminals will be exposed at each end, to a depth of a least 1 inch below the surface of the liquid (0001, 0002) Method 112, test condition A, MIL-STD-202 (0003).

Capacitance to ground: Method 305, MIL-STD-202.

Test Frequency: 1,000 ±100 Hz.

(D) Measured capacitance: Between 0.72 and 1.44 mFd. (0001 only).

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202. The test voltage shall be 800 V dc (0001 and 0002) and 2 times rated dc voltage (0003) applied between either terminal and case. The test voltage shall be applied and discharged through a resistance of at least 1 ohm per volt (0001 and 0002 only).

Barometric pressure (reduced): Method 105, MIL-STD-202 (0001 and 0002 only).

Test altitude: 80,000 feet (0.83 inch of mercury).

Dielectric withstanding voltage: In accordance with initial requirements, except the test voltage shall be 125 percent of the rated dc voltage.

Insulation resistance: Method 302, test condition A, MIL-STD-202.

Measurements at 25°C after test: Insulation resistance shall be at least 550 megohms (0001, 0002) and 1,000 megohm microfarads minimum (0003).

Voltage drop: 1.0 V rms, maximum (0001 and 0002) and .5 V dc, Max (0003).

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Insertion loss: In accordance with MIL-F-15733, and tables II and III (0001 and 0002); 60 dB at .15 MHz to 10 GHz (0003).

Overload: In accordance with MIL-F-15733.

Measurements at +25°C after test: Insulation resistance and voltage drop shall be measured and shall meet initial requirements.

Terminal strength: MIL-STD-202, method 211, test condition A (pull).

Applied force: 5 pounds. The force shall be applied in a direction parallel to the longitudinal axis of the filter. Slight deformation at the solder tab shall be allowable, provided no cracks, fractures or other damage to the terminal result from this test.

Flashpoint of impregnant or potting compound: In accordance with MIL-F-15733.

Minimum allowable flashpoint shall be +145°C.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, MIL-STD-202, test condition A, (0001 and 0002); test condition B (0003).

Thermal shock and immersion:

Thermal shock: Method 107, test condition A, MIL-STD-202.

Immersion: Method 104, test condition B, MIL-STD-202.

Measurements after final cycling:

Dielectric withstanding voltage: As initially specified, except the test voltage shall be 90 percent of that initially specified.

Insulation resistance: Not less than 30 percent of the initial value specified.

Insertion loss (check test) shall be measured and shall meet initial requirements.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Mounting: Filters shall be rigidly mounted by the body.

Electrical load: 100 percent of rated voltage between either terminal and case.

Insertion loss check test: Not applicable.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202. Electrical current load shall not be applied to filters during the vibration test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

After the vibration test, filters shall be subjected to and shall meet the initial requirements of the insertion-loss (check test).

Moisture resistance: Method 102, test condition D, MIL-STD-202, except no measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.

Initial measurements: Not applicable.

Polarization voltage: 100 V dc.

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Measurements after 24-hour drying period:

Dielectric withstanding voltage: As initially specified, except test voltage shall be 90 percent of that initially specified.

Insulation resistance shall be not less than 30 percent of the initial value specified.

Insertion-loss (check-test) shall meet initial requirements.

Life: Method 108, test condition B, MIL-STD-202. Test temperature +85°C.

Test voltage: 1.2 times the rated rms voltage.

Test current: 5 amperes, rms (0001 and 0002) and 2 amperes (0003) at 400 Hz.

Measurements after test:

Dielectric withstanding voltage: As initially specified, except the test voltage shall be 90 percent of that initially specified.

Insulation resistance shall be not less than 30 percent of the initial value specified.

Insertion-loss (check test) shall be measured and shall meet the initial requirements.

Part number: M15733/55- (dash no. from table I).

Supersession data:

This specification includes the requirements of contractor (13499), specification 241-0430-000, revision A, dated 28 May 1965, and contractor (86360) Drawing number 321C0650-01, dated 11 March 1970, and contractor (05869) Drawing 713880, Rev. B, dated 29 February 1972.

TABLE II. Insertion-loss (dB) (full load) versus frequency (MHz).

Type designation	Dash number	Minimum insertion loss full load in accordance with MIL-STD-220 at +25°C					
		0.15 MHz	0.3 MHz	0.6 MHz	1 MHz	10 MHz	20 MHz
FL85FW1EE3	0001	40	58	76	80	80	80
(D) FL85YW1EE3	0003	.15 MHz to 10 GHz					
				60			

TABLE III. Insertion-loss (dB) (no load) versus frequency (MHz).

Type designation	Dash number	Minimum insertion loss no load in accordance with MIL-STD-220 at +25°C									
		0.15 MHz	0.3 MHz	0.5 MHz	1 MHz	10 MHz	20 MHz	40-500 MHz	100 MHz	400 MHz	1000 MHz
FL85FW1EE3	0001	54	73	85	94	84	84	67	--	--	64
FL85FW1EE3	0002	54	--	85	94	84	--	--	80	80	--

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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